

QF620 Stochastic Modelling in Finance — Session 3

Reference Cheatsheet

Brownian Motion and Martingales: random walks, the definition and properties of W_t , the Markov property, geometric Brownian motion, Brownian martingales

Built from Session 3 slides (Tee Chyng Wen, SMU) · September 2026

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Source: “Session 3: Brownian Motion and Martingale” (Tee Chyng Wen, SMU, 25 slides). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged [*Beyond slides*]. $\triangle!$ marks a slide ambiguity or error, with the corrected version.

Exam context: the exam format is not stated on the Session 1–4 slides; check the course outline. From this session you need: the four defining properties of Brownian motion and how to test a candidate process against them; means, variances and covariances of W_t ; probabilities involving W_s, W_t jointly; the Markov property; moments of $e^{\sigma W_t}$ via the MGF; and the “add and subtract W_s ” technique for martingale proofs.

Notation used throughout (slide conventions)

Symbol	Meaning	Type	Notes
S_n	symmetric random walk after n steps	integer	starts at 0, ± 1 with probability $\frac{1}{2}$ [S2]
X_i	i.i.d. ± 1 steps	random	[S9]
$W_n(t)$	scaled random walk with n steps per unit time	random	steps of size $\sqrt{t/n}$ at spacing t/n [S9]
W_t	standard Brownian motion (Wiener process)	random, continuous	[S11]
Z	standard normal	random	
X_t	a candidate process, or GBM e^{W_t}	random	[S12], [S17]
$\mathbb{E}_s[\cdot], \mathbb{E}_t[\cdot]$	conditional expectation given information up to time s (or t)	random	[S16]
M_t	martingale	random	$M_t = \mathbb{E}_t[M_T]$ for all $T \geq t$ [S21]
$s < t$	two times	—	increments $W_t - W_s$
σ	real constant (exponent scale)	—	in $e^{\sigma W_t}$ [S24–S25]
$N(m, v)$	normal with mean m , variance v	—	

Notation trap: on [S2] S_4 is the random-walk position, not a stock price. In later sessions S_t is a stock.

0. Topic map

PREREQUISITES

S1 Normal distribution, MGF $E[e^{\theta X}] = e^{\{\mu\theta + \sigma^2\theta^2/2\}}$, lognormal moments
 S2 CLT; arithmetic tree $\pm\sigma\sqrt{T/n} \rightarrow$ normal; conditional expectation rules
 (linearity, known factors out, tower, independence); discrete martingales

↓

SESSION 3: BROWNIAN MOTION AND MARTINGALES

— BROWNIAN MOTION: HISTORY & RANDOM WALKS [S2–S10]

- | — Symmetric random walk: $2^4 = 16$ paths, range $-4..4$ [S2]
- | — Brown 1827, Einstein, Bachelier 1900 (thesis excerpts) [S3–S7]
- | — BM as the building block for continuous-time markets [S8]
- | — Scaled walk W_n : steps $\pm\sqrt{t/n}$ every t/n ; $W_n(t) = \sqrt{t} (\sum X_i/\sqrt{n})$ [S9]
- | — CLT $\Rightarrow W_n(t) \rightarrow N(0, t)$; independent, stationary increments [S10]

— DEFINITION OF W_t [S11–S15]

- | — $W_0 = 0$; continuous; stationary independent increments; $W_t - W_s \sim N(0, t-s)$ [S11]
- | — $\sqrt{t} \cdot Z$ is $N(0, t)$ at each t but NOT a BM [S12–S13]
- | — $E[W_t] = 0$, $E[W_t^2] = t$, $V[W_t - W_s] = t-s$, $\text{Cov}(W_s, W_t) = s$ ($s < t$); joint normal [S14]
- | — Examples: $P(W_2 < 0 \mid W_1 > 0)$, $P(W_1 W_2 > 0)$ [S15]

— PROPERTIES [S16–S20]

- | — Markov: $E_s[W_t] = W_s$; future depends only on the present [S16]
- | — GBM $X_t = e^{W_t}$; $E[X_t]$, $E[X_t^2]$ [S17]
- | — Continuous but nowhere differentiable; self-similar (fractal) [S18–S19]
- | — Reflection: $-W_t$ is also a BM [S20]

— MARTINGALES [S21–S25]

- | — $M_t = E_t[M_T]$; W_t is a martingale [S21]
- | — $W_t^2 - t$ is a martingale [S22]
- | — W_t^3 is not: $E_s[W_t^3] = W_s^3 + 3(t-s)W_s$ [S23]
- | — $E[e^{\sigma W_t}] = e^{\{\sigma^2 t/2\}}$ [S24]
- | — Exponential martingale $e^{\{\sigma W_t - \sigma^2 t/2\}}$ [S25]

↓

LATER: S4 stochastic integrals $\int f dW$ (mean 0 because increments are independent),

box calculus $(dW)^2 = dt$, Itô's formula – which re-derives every martingale here by checking for a zero dt term

Dependency logic:

- The random walk is the $\sigma = 1$ arithmetic tree of S2; scaling steps by $\sqrt{\Delta t}$ and the CLT give Brownian motion.
- Every computation this week reduces to one move: **write** $W_t = (W_t - W_s) + W_s$. The increment is independent of the past and $N(0, t - s)$; W_s is known at time s .
- Normal MGF (S1) $\Rightarrow \mathbb{E}[e^{\sigma W_t}] = e^{\frac{1}{2}\sigma^2 t} \Rightarrow$ GBM moments and the exponential martingale, which is the continuous-time version of “the discounted stock is a $\mathbb{Q}$ -martingale” (S2).

1 Concept entries

1.1 The symmetric random walk [S2]

- **Definition** [S2]: start at 0; each step $+1$ or -1 with probability $\frac{1}{2}$.
- **4-step example** [S2]: $2^4 = 16$ paths; maximum 4, minimum -4 .
- **Distribution** [Beyond slides]: with $B =$ number of up-steps $\sim \text{Bin}(4, \frac{1}{2})$, $S_4 = B - (4 - B) = 2B - 4$:

S_4	-4	-2	0	2	4
Probability	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

$\mathbb{E}[S_n] = 0$, $\mathbb{V}[S_n] = n$; after an even number of steps the walk is at an even value.

- **⚠** [S2] writes “ $S_4 \sim \text{Bin}(4, 0.5)$ ”. The binomial variable is the *number of up-steps*; the position is $S_4 = 2B - 4$ (it takes negative values, a binomial cannot).

1.2 History and motivation [S3–S8]

S3 : Robert Brown (1773–1858) observed that pollen particles’ motion came from the particles themselves, not from currents or evaporation.

S4 : Einstein showed these erratic movements obey a clear statistical law.

S5–S7 : Louis Bachelier, *Théorie de la spéculation* (1900): the market “unwittingly” obeys the law of probability. [S6–S7] reproduce pages of the thesis (stock-exchange operations; payoff diagrams of forward and option trades).

S8 : Bachelier proposed Brownian motion as a model of stock price fluctuations in 1900; it remains the building block of the reference continuous-time market model. Approach used here: BM as an infinitesimal random walk (smaller steps, more often).

- Not examinable in detail; know the chronology Brown $\rightarrow$ Bachelier/Einstein $\rightarrow$ Wiener (1923, rigorous construction) $\rightarrow$ Itô (1942, S4).

1.3 From random walk to Brownian motion [S9–S10]

- **Scaled random walk** [S9]: for a positive integer n :
 1. $W_n(0) = 0$.
 2. time spacing t/n .
 3. up and down jumps of equal size $\sqrt{t/n}$.
 4. probability $\frac{1}{2}$ each way (measure $\mathbb{P}$).

$$W_n\left(\frac{i}{n}t\right) = W_n\left(\frac{i-1}{n}t\right) + \sqrt{\frac{t}{n}} X_i, \quad W_n(t) = \sqrt{t} \left(\frac{\sum_{i=1}^n X_i}{\sqrt{n}} \right).$$

- **Why** $\sqrt{t/n}$ [S9]: with this scaling W_n does not blow up (or collapse) as n grows: $\mathbb{V}[W_n(t)] = n \cdot \frac{t}{n} = t$ for every n .
- **The limit** [S10]:
 - The bracket $\rightarrow N(0, 1)$ by the CLT, so $W_n(t) \rightarrow N(0, t)$.
 - Future moves away from any position are independent of where the walk is and of its history.
 - A displacement $W_n(s+t) - W_n(s)$ is (scaled) binomial with mean 0 and variance t , and tends to $N(0, t)$.
 - Brownian motion is the limit of the scaled random walk as $n \rightarrow \infty$.
- **Simulation** [Beyond slides] (tested): mean and variance are exactly 0 and t for every n ; only the *shape* becomes normal as n grows.

```
import numpy as np
rng = np.random.default_rng(0)
def scaled_walk(n, t, n_paths):
    X = rng.choice([-1, 1], size=(n_paths, n))
    return np.sqrt(t / n) * X.sum(axis=1) # W_n(t)
for n in (4, 100, 10_000):
    Wt = scaled_walk(n, 2.0, 20_000)
    print(n, Wt.mean().round(3), Wt.var().round(3)) # ~0, ~2 for every n
```

- **Connection:** this is S2's arithmetic tree with $\sigma = 1$; S2's $S_T = S_0 + \sigma\sqrt{T}Z$ becomes $S_0 + \sigma W_T$ (S4's first model).

1.4 Mathematical definition of Brownian motion [S11]

- **Definition** [S11] (Wiener, 1923): W_t is a Brownian motion iff
 1. $W_0 = 0$;
 2. W_t is continuous in $t \geq 0$ (no jumps);
 3. it has stationary and independent increments;
 4. for $0 \leq s \leq t$, $W_t - W_s \sim N(0, t - s)$.
 These are necessary and sufficient.
- **Reading the conditions** [Beyond slides]:
 - “Independent”: increments over non-overlapping intervals are independent; in particular $W_t - W_s$ is independent of everything up to time s .
 - “Stationary”: the distribution of $W_{t+h} - W_t$ depends only on h .
 - Condition 4 already implies stationarity; it is the *increment* condition that carries the content, not $W_t \sim N(0, t)$.
- **⚠** [S11] “Norbert Wiener” $\rightarrow$ Wiener.
- **Trap:** checking only $W_t \sim N(0, t)$ (marginal); see 1.5.

1.5 Testing a candidate: $X_t = \sqrt{t}Z$ is not a Brownian motion [S12–S13]

- **Slide example** [S12]: $Z \sim N(0, 1)$, $X_t = \sqrt{t}Z$. Continuous, and $X_t \sim N(0, t)$ for each t . Is it a BM?
- **Solution** [S12]: the increment uses the *same* Z :

$$X_{s+t} - X_s = (\sqrt{s+t} - \sqrt{s})Z \sim N\left(0, 2s + t - 2\sqrt{s(s+t)}\right).$$

- A BM increment over length t must have variance t . Fails.
- The increment is a multiple of X_s , so not independent of X_s . Fails.
- **Numbers** [Beyond slides]: $s = t = 1$: variance $3 - 2\sqrt{2} = 0.172$, not 1. $\text{Corr}(X_1, X_2) = 1$, whereas for BM $\text{Corr}(W_1, W_2) = 1/\sqrt{2}$.
- **Picture** [S13]: a BM path wiggles; $\sqrt{t}Z$ is a smooth $\sqrt{t}$ curve scaled by one random draw. Same marginals, completely different paths.
- **Lesson** [S13]: requiring normal increments of variance $t - s$, independent of the path up to s , is much stronger than requiring $W_t \sim N(0, t)$ for each t .
- **Checklist for “is X_t a BM?”** [Beyond slides]: (1) $X_0 = 0$? (2) continuous? (3) increments independent of the past? (4) $X_t - X_s \sim N(0, t - s)$? Common candidates: cW_{t/c^2} (yes), $-W_t$ (yes), $W_{t+h} - W_h$ (yes), $2W_{t/4}$ (yes), W_{2t} (no: variance $2t$), $\sqrt{t}Z$ (no).

1.6 Moments, covariance and joint normality [S14]

- **Properties** [S14]:

$$\mathbb{E}[W_t] = 0, \quad \mathbb{E}[W_t^2] = t, \quad \mathbb{E}[W_t - W_s] = 0, \quad \mathbb{V}[W_t - W_s] = t - s, \quad \text{Cov}(W_s, W_t) = s \ (s < t).$$

- **Derivation of the covariance** [Beyond slides]: $\mathbb{E}[W_s W_t] = \mathbb{E}[W_s(W_t - W_s)] + \mathbb{E}[W_s^2] = 0 + s$. In general $\text{Cov}(W_s, W_t) = \min(s, t)$; $\text{Corr} = \sqrt{s/t}$.
- **Joint normality** [S14]: for $t_1 < \dots < t_n$, $(W_{t_1}, \dots, W_{t_n})$ is multivariate normal with mean 0 and covariance

matrix $[\min(t_i, t_j)]$:

$$\begin{pmatrix} t_1 & t_1 & t_1 & \cdots & t_1 \\ t_1 & t_2 & t_2 & \cdots & t_2 \\ t_1 & t_2 & t_3 & \cdots & t_3 \\ \vdots & & & \ddots & \vdots \\ t_1 & t_2 & t_3 & \cdots & t_n \end{pmatrix}.$$

- **Higher moments** [Beyond slides]: $\mathbb{E}[W_t^3] = 0$, $\mathbb{E}[W_t^4] = 3t^2$.
- **Worked examples** [Beyond slides]:
 - $\mathbb{V}[W_1 + W_2] = 1 + 2 + 2(1) = 5$.
 - $\mathbb{V}[W_3 - 2W_1] = 3 + 4 - 4(1) = 3$. (Check: $W_3 - 2W_1 = (W_3 - W_1) - W_1$, variance $2 + 1 = 3$ ✓.)
 - $\mathbb{E}[W_2 W_5] = 2$; $\mathbb{E}[W_1 W_2 W_3] = 0$ (odd moment of a zero-mean normal vector).
- **Traps:** $\text{Cov}(W_s, W_t) = t$ or $= st$; forgetting the 2Cov term; treating W_s and W_t as independent (only their increments are).

1.7 Probability examples [S15]

- **Slide questions** [S15]: (a) conditional on $W_1 > 0$, $P(W_2 < 0)$? (b) $P(W_1 W_2 > 0)$?
- **Solution** [Beyond slides]: write $W_1 = X$, $W_2 = X + Y$ with X, Y i.i.d. $N(0, 1)$ (independent increments). (X, Y) is rotationally symmetric, so probabilities are angles/ 2π .
 - $\{W_1 > 0, W_2 < 0\} = \{X > 0, Y < -X\}$: a wedge of angle $\pi/4$, so probability $\frac{1}{8}$.
 - (a) $P(W_2 < 0 \mid W_1 > 0) = \frac{1/8}{1/2} = \frac{1}{4}$.
 - (b) By symmetry $P(W_1 < 0, W_2 > 0) = \frac{1}{8}$, so $P(W_1 W_2 > 0) = 1 - \frac{1}{4} = \frac{3}{4}$.
- **General formula** [Beyond slides]: for jointly normal with correlation ρ , $P(X > 0, Y > 0) = \frac{1}{4} + \frac{\arcsin \rho}{2\pi}$. Here $\rho = 1/\sqrt{2}$, $\arcsin \rho = \pi/4$, giving $\frac{3}{8}$ and $P(\text{same sign}) = \frac{3}{4}$ ✓.
- **Simulation check** [Beyond slides] (1,000,000 paths): 0.2501 and 0.7498.

```
t = np.array([1.0, 2.0])
dW = rng.normal(0, np.sqrt(np.diff(np.r_[0, t])), size=(1_000_000, 2))
W = dW.cumsum(axis=1) # columns W_1, W_2
print(np.cov(W.T).round(3)) # [[1, 1], [1, 2]]
print((W[W[:, 0] > 0, 1] < 0).mean(), (W[:, 0] * W[:, 1] > 0).mean()) # 0.25, 0.75
```

- **Trap:** answering $\frac{1}{2}$ for (a) by treating W_2 as independent of W_1 .

1.8 The Markov property [S16]

- **Definition** [S16]: the conditional distribution of the future depends only on the present state; all relevant information is in the present. Written as $\mathbb{E}_t[X_T] = \mathbb{E}[X_T \mid X_t, X_{t-1}, \dots, X_0] = \mathbb{E}[X_T \mid X_t]$.
- **BM is Markov** [S16]: for $0 < s < t$,

$$\mathbb{E}_s[W_t] = \mathbb{E}_s[W_t - W_s] + \mathbb{E}_s[W_s] = 0 + W_s.$$

To predict W_t you only need W_s .

- **Why** [S16]: every independent-increment process is Markov.
- **△!** [S16] states the Markov property with an expectation and demonstrates it with $\mathbb{E}_s[W_t] = W_s$. The Markov property concerns the whole *conditional distribution*: $W_t \mid \mathcal{F}_s \sim N(W_s, t - s)$, which depends on the past only through W_s [Beyond slides]. The identity $\mathbb{E}_s[W_t] = W_s$ is the *martingale* property (1.10). A process can be Markov without being a martingale (e.g. $W_t + t$) and vice versa [Beyond slides].

1.9 Geometric Brownian motion and path properties [S17–S20]

- **GBM** [S17]: $X_t = e^{W_t}$, a process whose logarithm is a Brownian motion; used for asset prices in Black–Scholes (with drift and volatility added: $S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$, S2 and S4).
- **Slide example** [S17] [Beyond slides] solution via the normal MGF with $W_T \sim N(0, T)$:
 - $\mathbb{E}[X_T] = \mathbb{E}[e^{W_T}] = e^{\frac{1}{2}T}$.
 - $\mathbb{E}[X_T^2] = \mathbb{E}[e^{2W_T}] = e^{\frac{1}{2} \cdot 4T} = e^{2T}$.
 - $\mathbb{V}[X_T] = e^{2T} - e^T$.
 - Since $\mathbb{E}[X_T] = e^{T/2} > 1 = X_0$, e^{W_t} is not a martingale; dividing by $e^{t/2}$ fixes it (1.12).

- **Path properties** [S18–S19]:
 - Continuous everywhere but differentiable nowhere (S4 builds on this).
 - Hits every real value eventually, however large or negative, and returns to 0 with probability 1.
 - Once it hits a value it immediately hits it again infinitely often, and keeps returning at arbitrarily large times.
 - Self-similar (fractal): looks the same at every scale.
- **Reflection** [S20]: $-W_t$ is also a Brownian motion (the normal distribution is symmetric about 0).
- **Other invariances** [Beyond slides]: scaling $\frac{1}{c}W_{c^2t}$; time shift $W_{t+h} - W_h$; time inversion $tW_{1/t}$ (with value 0 at $t = 0$). Scaling is the precise meaning of “fractal”.

1.10 Martingales; W_t and $W_t^2 - t$ [S21–S22]

- **Definition** [S21]: M_t is a martingale if $M_t = \mathbb{E}_t[M_T]$ for all $T \geq t$: the best forecast of the future value is today’s value. [Beyond slides] Technical conditions: M_t is known at time t (adapted) and $\mathbb{E}|M_t| < \infty$.
- **Standard proof technique**: write $W_t = (W_t - W_s) + W_s$, expand, and use (i) W_s is known at s , (ii) $W_t - W_s$ is independent of the past with $\mathbb{E} = 0$, $\mathbb{E}[(\cdot)^2] = t - s$, $\mathbb{E}[(\cdot)^3] = 0$.
- W_t is a martingale [S21]: $\mathbb{E}_s[W_t] = \mathbb{E}_s[W_t - W_s] + W_s = W_s$.
- $W_t^2 - t$ is a martingale [S22] ([Beyond slides] full expansion):

$$\mathbb{E}_s[(W_t - W_s + W_s)^2] = \underbrace{\mathbb{E}_s[(W_t - W_s)^2]}_{t-s} + 2W_s \underbrace{\mathbb{E}_s[W_t - W_s]}_0 + W_s^2 = W_s^2 + (t - s),$$

so $\mathbb{E}_s[W_t^2 - t] = W_s^2 - s$.

- **Consequence** [Beyond slides]: W_t^2 itself is not a martingale (it drifts up at rate 1); subtracting t (its quadratic variation, S4) removes the drift.
- **Traps**: forgetting the cross term; taking $\mathbb{E}_s[(W_t - W_s)^2] = t$ instead of $t - s$.

1.11 W_t^3 is not a martingale [S23]

- **Slide example** [S23]: $\mathbb{E}_s[(W_t - W_s + W_s)^3]$ expanded:

$$\mathbb{E}_s[(W_t - W_s)^3] + 3W_s \mathbb{E}_s[(W_t - W_s)^2] + 3W_s^2 \mathbb{E}_s[W_t - W_s] + W_s^3 = 0 + 3(t - s)W_s + 0 + W_s^3.$$

$\mathbb{E}_s[W_t^3] = W_s^3 + 3(t - s)W_s \neq W_s^3$: not a martingale.

- **Fixing it** [Beyond slides]: $W_t^3 - 3tW_t$ is a martingale, since $\mathbb{E}_s[W_t^3 - 3tW_t] = W_s^3 + 3(t - s)W_s - 3tW_s = W_s^3 - 3sW_s$. (Equivalently, via Itô, S4: $d(W^3 - 3tW) = (3W^2 - 3t)dW$, no dt term.)
- **⚠** [S23] concludes “ W_t^3 doesn’t satisfy the definition of a standard Brownian motion”; it should read “...the definition of a martingale”.

1.12 $\mathbb{E}[e^{\sigma W_t}]$ and the exponential martingale [S24–S25]

- **MGF recap** [S24]: $X \sim N(\mu, \sigma^2) \Rightarrow \mathbb{E}[e^{\theta X}] = e^{\mu\theta + \frac{1}{2}\sigma^2\theta^2}$; $Z \sim N(0, 1) \Rightarrow \mathbb{E}[e^{\theta Z}] = e^{\frac{1}{2}\theta^2}$.
- **Slide example** [S24]: $W_t \sim N(0, t)$, so with $\theta = \sigma$:

$$\mathbb{E}[e^{\sigma W_t}] = e^{\frac{1}{2}\sigma^2 t}.$$

- **Exponential martingale** [S25]: $M_t = \exp(\sigma W_t - \frac{1}{2}\sigma^2 t)$ is a martingale.

$$\mathbb{E}_s[e^{\sigma W_t - \frac{1}{2}\sigma^2 t}] = e^{\sigma W_s - \frac{1}{2}\sigma^2 s} \mathbb{E}_s[e^{\sigma(W_t - W_s)}] = e^{\sigma W_s - \frac{1}{2}\sigma^2 s} e^{\frac{1}{2}\sigma^2(t-s)} = e^{\sigma W_s - \frac{1}{2}\sigma^2 s}.$$

Steps: factor out the known $e^{\sigma W_s}$; the increment is independent and $N(0, t - s)$; apply the MGF.

- **Finance meaning** [Beyond slides]:
 - Risk-neutral GBM: $S_t = S_0 e^{(r - \frac{1}{2}\sigma^2)t + \sigma W_t}$, so $e^{-rt}S_t = S_0 M_t$: the discounted stock is a martingale, the continuous-time version of S2’s $M_n = S_n / (1 + r)^n$.
 - M_t is also the Radon–Nikodym density used to change drift (Girsanov), the continuous analogue of S2’s dQ/dP .
- **Simulation check** [Beyond slides]: $\sigma = 0.5$, $t = 2$: mean of $e^{\sigma W_2 - \frac{1}{2}\sigma^2 \cdot 2}$ over 10^6 paths = 1.0002; mean of $e^{W_2} = 2.719$ vs $e^1 = 2.718$.
- **Traps**: $\mathbb{E}[e^{\sigma W_t}] = 1$ (forgetting convexity); $e^{\frac{1}{2}\sigma^2 t}$ (variance is $\sigma^2 t$).

2 Comparison tables

2.1 Random walk vs scaled walk vs Brownian motion

	Symmetric walk [S2]	Scaled walk W_n [S9]	Brownian motion [S11]
Time	integer steps	spacing t/n	continuous
Step size	± 1	$\pm \sqrt{t/n}$	infinitesimal
Distribution at t	$2 \text{Bin}(n, \frac{1}{2}) - n$	scaled binomial, variance t	$N(0, t)$
Increments	independent	independent	independent, $N(0, t - s)$
Paths	jumps	jumps	continuous, nowhere differentiable

2.2 Is it a Brownian motion?

Process	Reason	BM?
W_t	definition	yes
$-W_t$ [S20]	symmetry of the normal	yes
$\frac{1}{c}W_{c^2t}$ [Beyond slides]	increments $N(0, t - s)$, independent	yes
$W_{t+h} - W_h$ [Beyond slides]	fresh start	yes
$\sqrt{t}Z$ [S12]	increment variance wrong; not independent	no
W_{2t} [Beyond slides]	increment variance $2(t - s)$	no
$W_t + t$ [Beyond slides]	increment mean $t - s \neq 0$	no

2.3 Martingale or not? [S21–S25]

Process	$\mathbb{E}_s[\cdot]$	Martingale?
W_t	W_s	yes [S21]
W_t^2	$W_s^2 + (t - s)$	no
$W_t^2 - t$	$W_s^2 - s$	yes [S22]
W_t^3	$W_s^3 + 3(t - s)W_s$	no [S23]
$W_t^3 - 3tW_t$ [Beyond slides]	$W_s^3 - 3sW_s$	yes
e^{W_t}	$e^{W_s} e^{\frac{1}{2}(t-s)}$	no [S17]
$e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$	$e^{\sigma W_s - \frac{1}{2}\sigma^2 s}$	yes [S25]

2.4 Markov vs martingale [Beyond slides]

	Markov [S16]	Martingale [S21]
Statement	future <i>distribution</i> depends only on the present	future <i>expectation</i> equals the present
BM	yes: $W_t \mathcal{F}_s \sim N(W_s, t - s)$	yes: $\mathbb{E}_s[W_t] = W_s$
$W_t + t$	yes	no
$W_t^2 - t$	yes	yes

3 Master formula sheet

Random walks [S2, S9–S10]

- $S_n = 2B - n$, $B \sim \text{Bin}(n, \frac{1}{2})$; $\mathbb{E}[S_n] = 0$, $\mathbb{V}[S_n] = n$.
- $W_n(t) = \sqrt{t} \frac{1}{\sqrt{n}} \sum X_i \rightarrow N(0, t)$.

Definition [S11]: $W_0 = 0$; continuous; independent stationary increments; $W_t - W_s \sim N(0, t - s)$. **Moments [S14]**

- $\mathbb{E}[W_t] = 0$, $\mathbb{E}[W_t^2] = t$, $\mathbb{E}[W_t^4] = 3t^2$ [Beyond slides]; $\mathbb{V}[W_t - W_s] = t - s$.
- $\text{Cov}(W_s, W_t) = \min(s, t)$; joint covariance matrix $[\min(t_i, t_j)]$.
- $P(W_s > 0, W_t > 0) = \frac{1}{4} + \frac{\arcsin \sqrt{s/t}}{2\pi}$ [Beyond slides].

Conditional expectations ($s < t$)

- $\mathbb{E}_s[W_t] = W_s$; $\mathbb{E}_s[W_t^2] = W_s^2 + (t - s)$; $\mathbb{E}_s[W_t^3] = W_s^3 + 3(t - s)W_s$.
- $\mathbb{E}_s[e^{\sigma W_t}] = e^{\sigma W_s + \frac{1}{2}\sigma^2(t-s)}$.

Exponentials [S17, S24–S25]

- $\mathbb{E}[e^{\sigma W_t}] = e^{\frac{1}{2}\sigma^2 t}$; $\mathbb{E}[e^{W_T}] = e^{T/2}$; $\mathbb{E}[e^{2W_T}] = e^{2T}$.
- $e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$ is a martingale.

Martingales [S21–S23]: W_t , $W_t^2 - t$, $W_t^3 - 3tW_t$ [Beyond slides], $e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$.

4 Glossary

- **Adapted:** known at time t given the history up to t .
- **Brownian motion (Wiener process):** $W_0 = 0$, continuous, independent stationary increments, $W_t - W_s \sim N(0, t - s)$ [S11].
- **Exponential martingale:** $e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$ [S25].
- **Fractal / self-similar:** statistically the same at every scale [S19].
- **Geometric Brownian motion:** a process whose log is a (drifted, scaled) BM, e.g. e^{W_t} [S17].
- **Increment:** $W_t - W_s$.
- **Independent increments:** increments over disjoint intervals are independent [S11].
- **Markov process:** future distribution depends only on the present state [S16].
- **Martingale:** $M_t = \mathbb{E}_t[M_T]$ for $T \geq t$ [S21].
- **Nowhere differentiable:** no derivative at any time [S18–S19].
- **Reflection:** $-W_t$ is a BM [S20].
- **Scaled random walk:** $\pm\sqrt{t/n}$ steps every t/n [S9].
- **Stationary increments:** increment distribution depends only on the length of the interval [S11].
- **Symmetric random walk:** ± 1 steps with probability $\frac{1}{2}$ [S2].

5 Self-test

Questions

- Q1 ([S2]).** For the 4-step symmetric random walk, give the distribution of S_4 and its mean and variance. Why is “ $S_4 \sim \text{Bin}(4, 0.5)$ ” not literally correct?
- Q2.** Explain why the scaled walk uses steps $\pm\sqrt{t/n}$ and show $\mathbb{V}[W_n(t)] = t$ for every n . What does the CLT add?
- Q3.** State the four defining properties of Brownian motion.
- Q4 ([S12]).** Show that $X_t = \sqrt{t}Z$ is not a Brownian motion. Compute $\mathbb{V}[X_2 - X_1]$ and $\text{Corr}(X_1, X_2)$ and compare with BM.
- Q5.** Find $\text{Cov}(W_2, W_5)$, $\mathbb{V}[W_1 + W_2]$, $\mathbb{V}[W_3 - 2W_1]$ and $\text{Corr}(W_1, W_4)$.
- Q6 ([S15]).** Find $P(W_2 < 0 \mid W_1 > 0)$ and $P(W_1 W_2 > 0)$.
- Q7 ([S16]).** Show $\mathbb{E}_s[W_t] = W_s$. Is this the Markov property? Explain the difference between Markov and martingale.
- Q8 ([S17]).** $X_t = e^{W_t}$. Find $\mathbb{E}[X_T]$, $\mathbb{E}[X_T^2]$ and $\mathbb{V}[X_T]$. Is X_t a martingale?
- Q9 ([S22–S23]).** Show $W_t^2 - t$ is a martingale and W_t^3 is not. Find c such that $W_t^3 - ctW_t$ is a martingale.
- Q10 ([S25]).** Show $\exp(\sigma W_t - \frac{1}{2}\sigma^2 t)$ is a martingale, and explain its link to risk-neutral pricing.

Solutions

S1. $S_4 \in \{-4, -2, 0, 2, 4\}$ with probabilities $\frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}$; $\mathbb{E} = 0$, $\mathbb{V} = 4$. $S_4 = 2B - 4$ where $B \sim \text{Bin}(4, \frac{1}{2})$ counts up-steps; a binomial cannot be negative.

S2. Each step has variance t/n ; n independent steps give t . Smaller steps would make the limit degenerate, larger ones would explode. The CLT makes the limiting *distribution* normal: $N(0, t)$.

S3. $W_0 = 0$; continuous paths; independent, stationary increments; $W_t - W_s \sim N(0, t - s)$.

S4. $X_2 - X_1 = (\sqrt{2} - 1)Z$, variance $(\sqrt{2} - 1)^2 = 3 - 2\sqrt{2} = 0.172 \neq 1$, and it is a multiple of X_1 (not independent). $\text{Corr}(X_1, X_2) = 1$ vs $1/\sqrt{2}$ for BM.

S5. $\text{Cov}(W_2, W_5) = 2$; $\mathbb{V}[W_1 + W_2] = 1 + 2 + 2 = 5$; $\mathbb{V}[W_3 - 2W_1] = 3 + 4 - 4 = 3$; $\text{Corr}(W_1, W_4) = 1/\sqrt{4} = \frac{1}{2}$.

S6. $W_1 = X$, $W_2 = X + Y$; $P(X > 0, Y < -X) = \frac{1}{8}$ (angle $\pi/4$). $P(W_2 < 0 \mid W_1 > 0) = \frac{1}{4}$; $P(W_1 W_2 > 0) = 1 - 2 \cdot \frac{1}{8} = \frac{3}{4}$.

S7. $\mathbb{E}_s[W_t] = \mathbb{E}_s[W_t - W_s] + W_s = W_s$. This is the martingale property. Markov is the statement that the whole conditional distribution, $N(W_s, t - s)$, depends on the past only through W_s . $W_t + t$ is Markov but not a martingale.

S8. $\mathbb{E}[e^{W_t}] = e^{t/2}$; $\mathbb{E}[e^{2W_t}] = e^{2t}$; $\mathbb{V} = e^{2t} - e^t$. Not a martingale: $\mathbb{E}_s[e^{W_t}] = e^{W_s} e^{(t-s)/2} \neq e^{W_s}$.

S9. $\mathbb{E}_s[W_t^2] = W_s^2 + (t - s) \Rightarrow \mathbb{E}_s[W_t^2 - t] = W_s^2 - s$. $\mathbb{E}_s[W_t^3] = W_s^3 + 3(t - s)W_s \neq W_s^3$. $\mathbb{E}_s[W_t^3 - ctW_t] = W_s^3 + 3(t - s)W_s - ctW_s$; need $3(t - s) - ct = -cs$ for all t , so $c = 3$.

S10. See 1.12. With $S_t = S_0 e^{(r - \frac{1}{2}\sigma^2)t + \sigma W_t}$, the discounted price $e^{-rt} S_t = S_0 e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$ is a martingale: the continuous-time version of $S_n/(1 + r)^n$ being a $\mathbb{Q}$ -martingale, so $S_0 = \mathbb{E}^\mathbb{Q}[e^{-rT} S_T]$.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title	header
S2	Symmetric random walk, 4 steps	1.1
S3	Robert Brown quote	1.2
S4	C. P. Snow quote on Einstein	1.2
S5	Bachelier quote (with translation)	1.2
S6	Bachelier thesis excerpt: stock-exchange operations	1.2
S7	Bachelier thesis excerpt: payoff diagrams	1.2
S8	BM as a stock model; infinitesimal random walk; sample path	1.2, 1.3
S9	Scaled random walk definition	1.3
S10	CLT limit; BM as limit of scaled walks	1.3
S11	Definition of Brownian motion	1.4
S12	Example: $\sqrt{t}Z$ is not a BM	1.5
S13	Marginal vs increment conditions; path comparison	1.5
S14	Moments; covariance; joint normal covariance matrix	1.6
S15	Examples: $P(W_2 < 0 \mid W_1 > 0)$, $P(W_1 W_2 > 0)$	1.7
S16	Markov property	1.8
S17	Geometric Brownian motion; $\mathbb{E}[X_T]$, $\mathbb{E}[X_T^2]$	1.9
S18	Nowhere differentiable, self-similar (picture)	1.9
S19	Fractal and recurrence properties	1.9
S20	Reflection invariance	1.9
S21	Martingale definition; W_t is a martingale	1.10
S22	$W_t^2 - t$ is a martingale	1.10
S23	W_t^3 is not a martingale	1.11
S24	MGF recap; $\mathbb{E}[e^{\sigma W_t}]$	1.12

Slide	Content	Section
S25	Exponential martingale	1.12

Appendix B. Flag summary ()

1. [S2] “ $S_4 \sim \text{Bin}(4, 0.5)$ ”: the number of up-steps is binomial; the position is $2B - 4$.
2. [S11] “Weiner” $\rightarrow$ Wiener.
3. [S16] Markov property stated via expectations and demonstrated with $\mathbb{E}_s[W_t] = W_s$, which is the martingale property; Markov is about the full conditional distribution.
4. [S21] Martingale definition omits adaptedness and integrability ($\mathbb{E}|M_t| < \infty$).
5. [S23] “doesn’t satisfy the definition of a standard Brownian motion” $\rightarrow$ “of a martingale”.
6. [S15, S17, S24] Examples posed without solutions; the answers here ($\frac{1}{4}$, $\frac{3}{4}$, $e^{T/2}$, e^{2T} , $e^{\frac{1}{2}\sigma^2 t}$) are my own and are checked by simulation.